

When Faces Are Masked: Exploring Emotional Expression Through Body Postures in Virtual Reality

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ABSTRACT

As simulation advances in healthcare training, understanding how body-only signals convey emotions in virtual environments is crucial, particularly with masked virtual agents. This study involved 41 nursing students evaluating 16 faceless fear and surprise postures to assess their realism and the emotion conveyed. While well-recognized in 2D human representations, only three of 16 postures were correctly identified by more than 50% of participants in a 3D virtual agent. These results highlight the impact of virtual agent design on emotional recognition and the need for rigorous testing and refinement to improve emotional expressiveness and realism.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Virtual Reality—Empirical studies in HCI

1 INTRODUCTION

Simulation is a key tool for team training in high-stakes environments [2]. Effective teamwork relies on emotional communication, fostering collaboration, especially during crises [7]. Virtual agents in simulation often use facial expression as the primary channel for communication conveying emotions [9]. Studies show that virtual characters' emotional expressions are better recognized when combining facial and body cues [5]. However, in operating room teams, masks hinder facial expression, making the body the main expressive channel [4]. This study aims to simulate faceless emotional postures using 3D masked virtual agents, assessing their recognition and perception by nursing students.

2 METHOD

2.1 Participants

We recruited 41 nursing students ($Age = 23.18$, $SD = 6.46$), 93% of whom were female. Among participants, 63% had prior experience with virtual reality while 73% indicated they either never engage in video gaming or do so only a few times per year.

2.2 Procedure

After providing informed consent, participants completed a pre-questionnaire measuring their anxiety levels followed by the virtual

reality (VR) phase where they observed an artificial agent adopting various emotional postures. For each posture, they identified the expressed emotion and rated its realism. Finally, after viewing all stimuli, participants completed post-questionnaires to assess social presence and their impressions of the virtual agent.

2.3 Material

We developed a virtual operating room in Unity, displayed via an HTC Vive Pro, featuring a virtual agent representing an orthopedic surgeon in full surgical attire, including a helmet, hood, gown, and mask. The emotional postures were derived from the BESST dataset [11] which include 565 frontal view images of real bodies with blurred facial expressions depicting six emotions (happiness, sadness, fear, anger, disgust, and surprise). We focused on fear and surprise body postures and selected the 16 most accurately recognized and highly rated for realism from the original article data. These postures were recreated on a 3D model using a motion capture system (Xsens motion capture suit by Movella) and integrated into the operating room in Unity.

2.4 Measures

First, participants' pre-VR anxiety levels were measured using the 6-item State-Trait Anxiety Inventory [8] on a 4-point Likert scale (1 = "Not at all", 4 = "Very much"). Then, participants evaluated the virtual agent's emotional expressiveness by identifying emotions conveyed by each posture, choosing from six options: fear, sadness, surprise, happiness, anger, and disgust (Figure 1). They also rated each posture's realism on a 7-point Likert scale ranging from "Not at all realistic" to "Completely realistic".

After viewing the 16 postures, social presence was assessed using a 5-item scale developed by Bailenson and colleagues [1], with responses on a 5-point Likert scale from 1 ("Not at all") to 5 ("Totally"). An example item is: "The idea that the person isn't a real person has often crossed my mind." Perception of the virtual agent was qualitatively assessed using three items from Ho and McDorman's uncanny valley scale [6], evaluating attractiveness (repulsive, agreeable), eeriness (ordinary, weird), and humanness (natural, real) on a 7-point Likert scale.

3 RESULTS

The results showed that only three out of the 16 postures were correctly identified by more than 50% of participants, all of which were fear postures. Postures A was recognized by 76% of participant, posture B by 58% (Figure 2), and the third posture by 51% only.

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Figure 1: Emotion recognition task interface



Figure 2: Fear Body Posture A (left panel) and B (right panel)

In contrast, the highest recognition rate for surprise postures was 49%, suggesting that surprise is more difficult to convey through body-only cues. Interestingly, among the two most recognized fear postures, a difference in perceived realism emerged: Posture A was rated as more realistic ($M = 5.17, SD = 1.73$) than posture B ($M = 3.97, SD = 1.68$). The perception of the virtual agent was ambivalent, being rated as synthetic ($M = 3.22, SD = 1.86$) and ordinary ($M = 3.68, SD = 1.69$) with a neutral judgment on repulsiveness ($M = 3.98, SD = 1.30$). Social presence was moderate ($M = 2.57, SD = 1.30$), suggesting the interaction did not create a fully immersive or credible social experience. Social presence was positively correlated with perceiving the virtual agent as more agreeable ($r = .31, p < .05$), ordinary ($r = .52, p < .001$), and real ($r = .61, p < .001$). Anxiety showed no correlation with the emotion recognition, but correct identification was negatively correlated with perceiving the virtual agent as more real than synthetic ($r = -.33, p < .05$).

4 CONCLUSION

This study highlights the challenges of recognizing emotions through body-only cues on 3D virtual agents. Despite selecting 16 highly recognizable fear and surprise postures from a validated dataset [11], recognition rates were lower on virtual agents, suggesting that additional factors may impact emotional interpretation in virtual environments. One possible explanation is that animation complexity in virtual agents may hinder emotion recognition by introducing subtle movements that complicate interpretation. In addition, the uncanny valley effect could contribute [6], as participants perceived the virtual agent as synthetic, limiting participants' ability to connect with the virtual agent and accurately interpret its emotional expressions. This is supported by the virtual agent's lack of credibility needed to establish a strong sense of social presence.

Although fear and surprise postures had similar recognition rates in 2D images, surprise was harder to recognize in 3D, possibly due to the complexity of surprise. Unlike fear, which is a basic emotion

with universal and recognizable cues, surprise may be a context-dependent emotion or mental state [10] and may require more subtle visual information to be conveyed [3]. Prior research [5] also highlighted that surprise relies heavily on facial expressions, supporting the idea that body-only cues are insufficient.

This study has limitations: while frontal postures were selected from the original dataset, participants in the virtual reality environment viewed the virtual agent from a three-quarter profile perspective. Also, no multimodal cues (voice or text) were used. Emotional recognition is inherently multimodal, and the absence of these signals may have reduced recognition accuracy. Future studies should integrate other cues to enhance emotional conveyance. These findings emphasize the need for a VR-specific body-language lexicon, as body-only emotional communication is inherently more challenging to convey in virtual environments. Improving this aspect is crucial for applications such as collaborative virtual environments and medical training scenarios where facial cues are masked. Future research should focus on improving the design of virtual agents to mitigate the uncanny valley effect and incorporate voice-based interactions or dynamic gestures to strengthen the social presence of virtual agents.

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